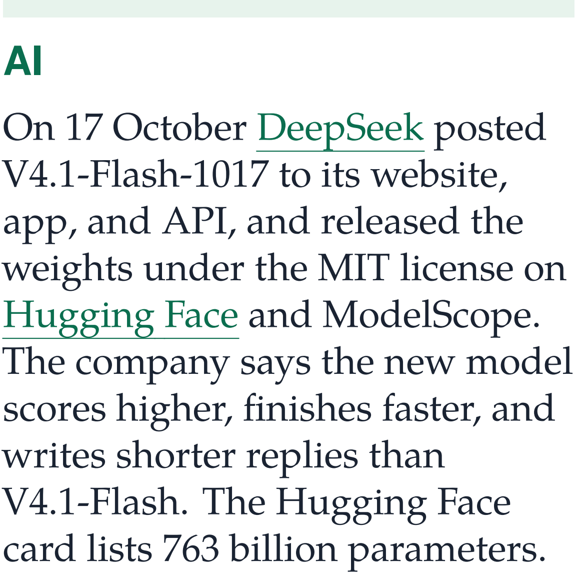
Dave's Daily

tldr

1. On 17 October DeepSeek published V4.1-Flash-1017 and released the weights. On its DeepSWE v1.1 table the score is 77.0, against 74.2 for the Flash model it replaces.
2. Reuters reported on 16 October that CoreWeave bonds issued in June at a 9.675 percent yield were yielding 15 percent. A second Reuters report the same day put the debt load near $50 billion.
3. FGSS-BO stores a language model's search idea as a record a program can test. On a cell-protocol task, six paired runs, the mean best score was 0.813, against 0.691 without that record.

AI

On 17 October DeepSeek posted V4.1-Flash-1017 to its website, app, and API, and released the weights under the MIT license on Hugging Face and ModelScope. The company says the new model scores higher, finishes faster, and writes shorter replies than V4.1-Flash. The Hugging Face card lists 763 billion parameters. The API name stays `deepseek-flash`. The retired names `deepseek-v4-flash` and `deepseek-v4-flash-vision-exp` still connect, and those calls are billed at the Flash price.



Score on five tests, previous V4.1-Flash in gray, V4.1-Flash-1017 in green. Higher is better. The axis runs from 0 to 100. Terminal-Bench 4.0 uses the Claude Code harness. DeepSWE v1.1 uses Mini-SWE-V2. Source: DeepSeek's Hugging Face card, 17 October 2026.

The card is the vendor's. Opus 5 still leads it on Humanity's Last Exam, 56.3 against 42.6, and on ProgramBench, 37.0 against 29.1. Flash 1017 leads Opus 5 on DeepSWE (74.0), Automation-Bench (50.3), Agents' Last Exam (28.6), and Apex Bench (65.8). The largest move in the figure is Terminal-Bench, from 31.2 to 42.3. The same post opens Search and Fetch at $0.75 per 1,000 calls. DeepSeek says the engine serves hundreds of millions of requests a day and filters search-engine manipulation, generative-engine manipulation, and low-quality generated text. Calls through DeepSeek Harness, Claude Code, and Codex stay free until 31 October 2026 at 16:00 UTC.

Markets

On 16 October Suzanne McGee and Gertrude Chavez-Dreyfuss reported for Reuters that CoreWeave bonds issued in June at a yield of 9.675 percent were yielding 15 percent. They treat the bond move as the sharper test. The debt is how the data centers get built, and the buildings may take years to earn revenue. Shashwat Chauhan and Niket Nishant, in a second Reuters report the same day, put the debt load at about $50 billion. It says the yield on the October 2031 dollar bond briefly passed 16 percent on Thursday, and that the cost of insuring the debt rose past 1,000 basis points, a record for this borrower, citing S&P Global Market Intelligence. One thousand basis points is 10 percentage points a year for that insurance.

Larry Holzenthaler, a Catalyst portfolio manager quoted by McGee and Chavez-Dreyfuss, says the risks now being priced were already visible in June: the size of the funding, the leverage, the dependence on outside money, and the chance that the graphics chips are outdated before the buildings open. Jim Reid of Deutsche Bank, in a note the same story cites, separates the center from the edge. AI financing costs are not dramatic overall. There are worrying themes at the edges. The yield is the price of waiting to see which buildings open.

Bayesian optimization

arXiv:2610.21246, submitted 15 October, is *FGSS-BO: Feedback-Guided Hypothesis Management for LLM-Based Bayesian Optimization*, by Hao Chen, Ziyang Ling, Ronald X. Xu, and Mingzhai Sun (University of Science and Technology of China).

LLAMBO already asks a language model to propose configurations and to predict scores. It does not keep a named guess about which setting matters. A suggestion to lower one knob can arrive bundled with changes to others, so the measured result does not test the suggestion.

FGSS stores each guess as a record: an identifier, the parameters it touches, an anchor configuration, the change to try, the direction it expects, and the evidence it cites. A reflector, which is a language-model call, writes and revises the records. A deterministic curator checks them, merges duplicates, and caps how many stay active. The surrogate stays LLAMBO's. It uses the language model's predicted mean and standard deviation, and the default choice is expected improvement. Once a round, the search may spend the evaluation on a matched test instead. A matched test changes one or two variables against the anchor. Only that comparison updates the record's support.

The primary experiments do not run a new laboratory. They look up a nearest neighbor in an old table. Ten low-scoring starts, then twenty searches, one at a time, six paired runs. The model is GPT-4o mini. Each reflection is one call at temperature zero, capped at 1,500 tokens.

On the iPSC-RPE protocol task, mean final best was 0.813 (sample standard deviation 0.023) for FGSS-BO, 0.691 (0.117) for the matched LLAMBO, 0.793 (0.026) for Gaussian-process expected improvement, 0.780 (0.078) for random search, and 0.569 (0.169) for HEBO. Search AUC, the average of the best-so-far after the start, was 0.640 against 0.494 for LLAMBO. The paired interval versus the Gaussian process includes zero for the final best and sits above zero for the AUC. Six runs is a small sample.

The pieces do not all pull the same way. On iPSC, a prose summary with no hypothesis cards already reaches a mean final best of 0.785. Full FGSS is 0.0285 higher on that score and 0.0061 lower on search AUC. Always choosing by expected improvement, while still keeping the cards, gives search AUC 0.690, above full FGSS. The paired intervals on those gaps cross zero.

Where the objective is a model trained from scratch, the clearest change is a diabetes support-vector machine. Mean cross-validated mean squared error falls from 0.57768 to 0.51711, which the paper states as 10.48 percent, and search AUC falls from 0.61046 to 0.55636. Lower is better on that task. Other hyperparameter tasks are mixed. On five chemical-reaction tasks, HEBO and the Gaussian process beat both language-model methods on the sample means. On an exact grid of the iPSC table, with no nearest-neighbor fill-in, the mean final best is close, 0.840 for LLAMBO and 0.848 for FGSS-BO, and mean search AUC favors LLAMBO, 0.750 against 0.710.

On iPSC, before a feasible-candidate fix, 18.4 percent of raw proposals were unique and 3.4 percent passed validity and novelty checks.

Updated 18 October 2026. Phone and letter match. Sources: DeepSeek and Hugging Face, 17 October; Reuters, 16 October; arXiv, 15 October. The bond yields are secondary-market prices, not a forecast. Created with malvin.